

Natural Language Processing : Basics to Advance Concept and Application

Vijay Dwivedi



Natural Language Processing

What is NLP?

By “natural language” we mean a language that is used for everyday communication by humans.

NLP is an Intersection of several fields

- Computer Science
- Artificial Intelligence
- Linguistics
- Statistics

It is basically teaching computers to process human language

Two main components:

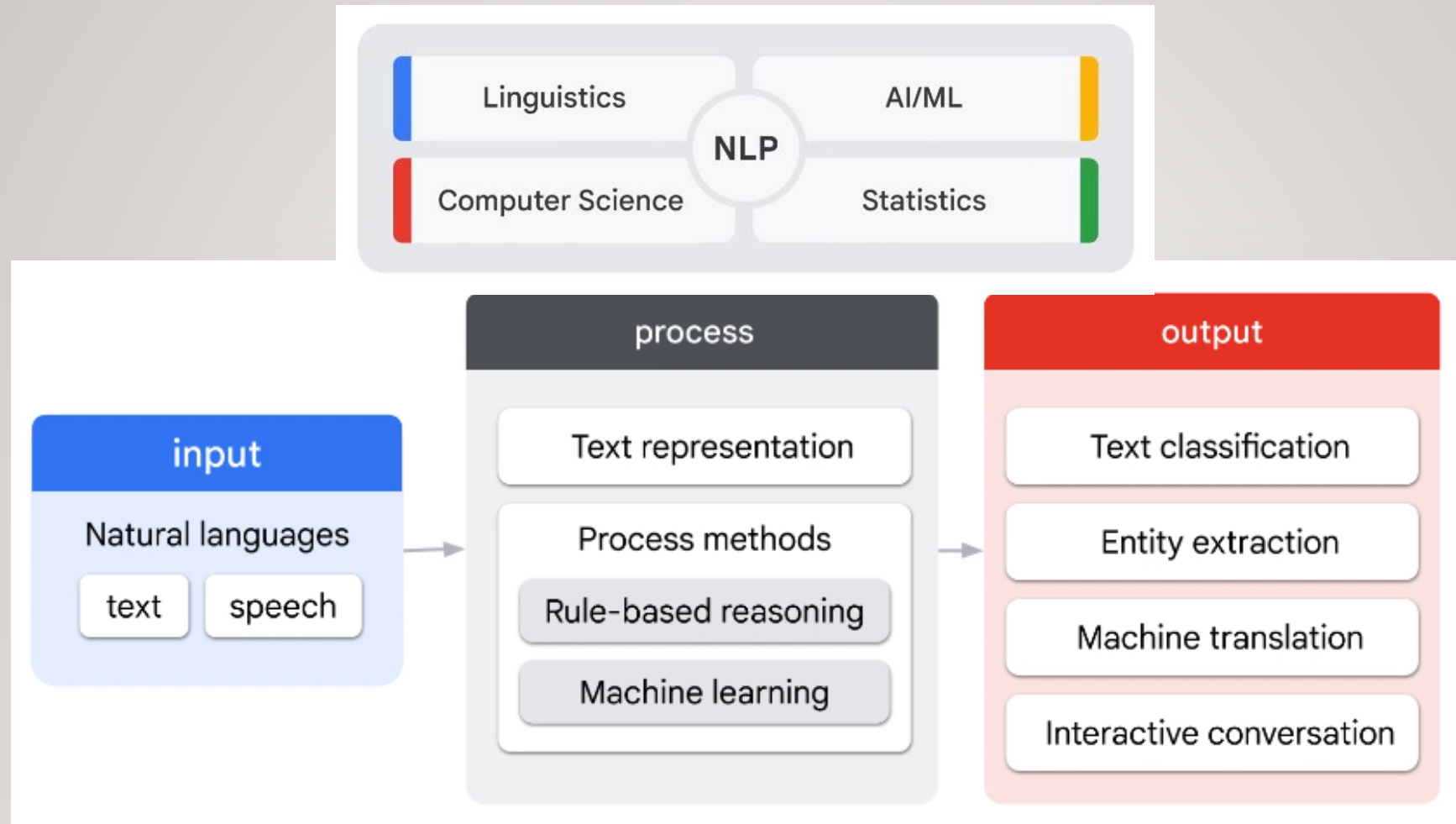
- Natural Language Understanding (NLU)
- Natural Language Generation (NLG)

NLP is AI Complete

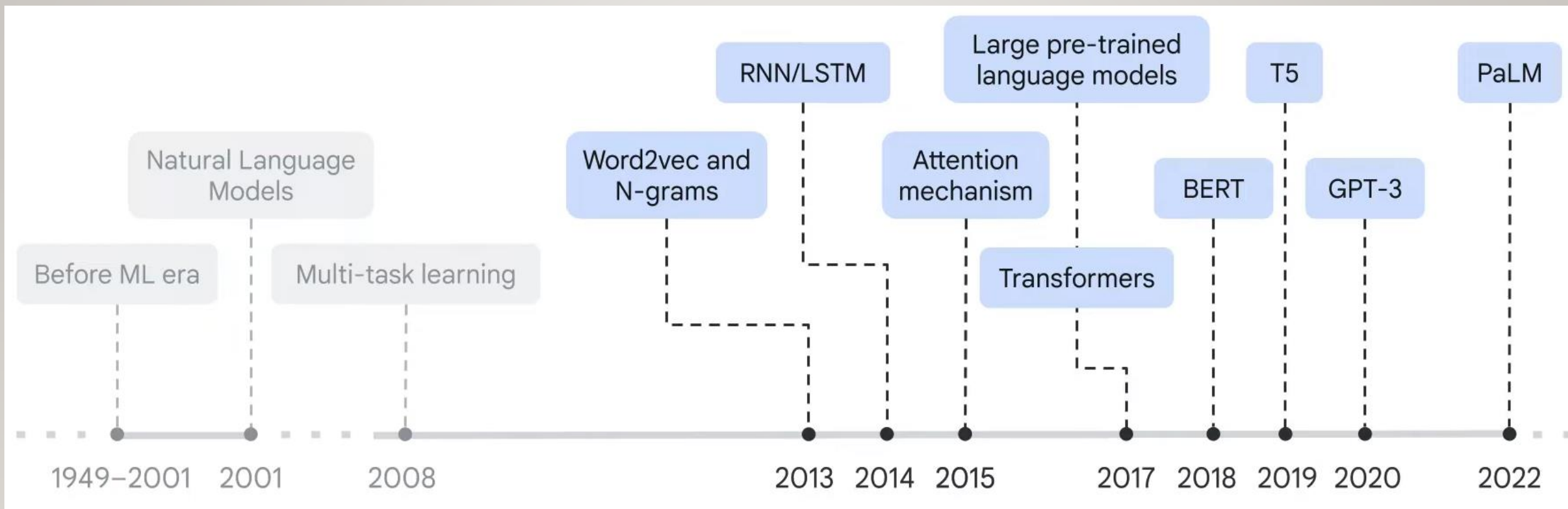
- Requires all types of knowledge humans possess → It's hard!



Natural Language Processing



Natural Language Processing : Language Model History



Natural Language Processing

NLU Applications

ML on Text (Classification, Regression, Clustering)

Classify emails as spam vs. not spam

Document Recommendation

Recommend news articles based on past articles liked

Sentiment Analysis

Extracting Word/Document Meaning (vectors) and

Relationship Extraction

Topic Modeling

Find documents belonging to a certain topic



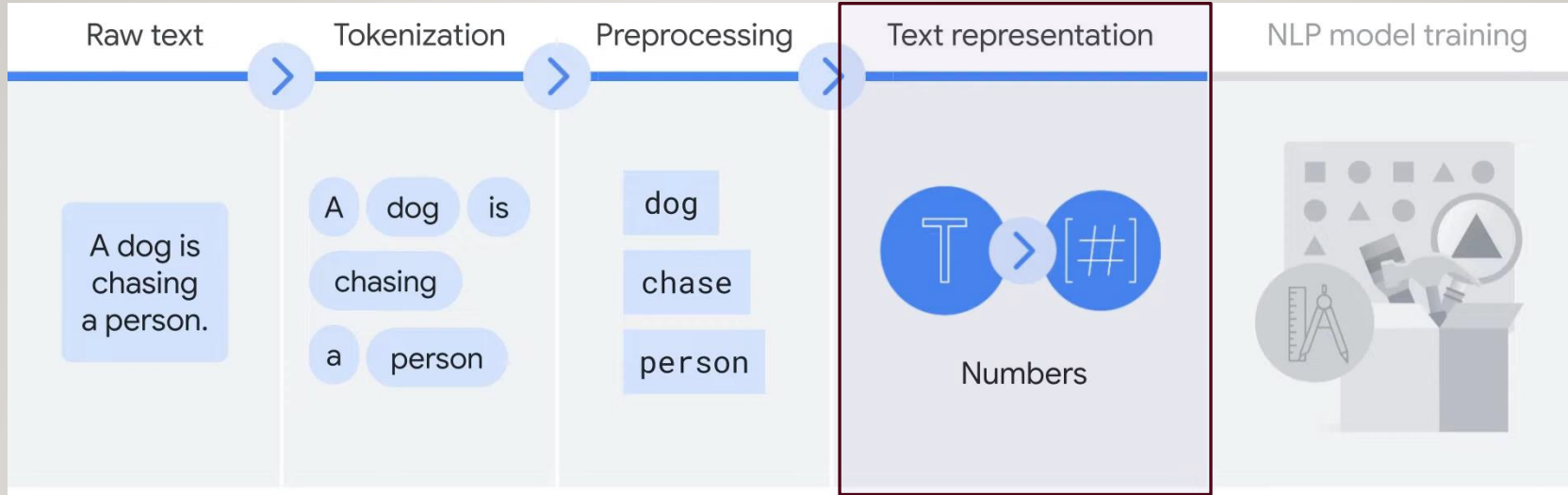
Natural Language Processing

NLG Applications

- Image Captioning
- Text Summarization
- Machine Translation
- Question Answering/Chatbots
- Notice NLU is almost a prerequisite for NLG



Text Representation – Feature Engineering



Text Representation:

1. Basic Vectorization :
 - ❖ One hot encoding & Bag of word
2. Word embeddings
 - ❖ Word2vec is family of model architecture and optimizations that can be used to learn word embeddings from large datasets (CBOW, Skip gram)
3. Transfer Learning

Word Embeddings

- Embeddings are a way to encode words capturing the **context** around them.
- Bag of Words -> Sparse one-hot encoded vectors; Count; TF-IDF; Co-occurrence matrix
- Idea of word embeddings:
 - Dimension Reduction
 - Contextual Similarity
 - Semantic Information of words and relations between them

Word Embeddings

- Dimension Reduction
 - If the vocabulary is of 50,000 words, and if the document has a maximum of 500 words, then the document is represented by $50000 * 500$ matrix
 - However, with word embeddings, imagine we assume 20 vector length for a word, then the embedding matrix would be of $50000 * 20$.

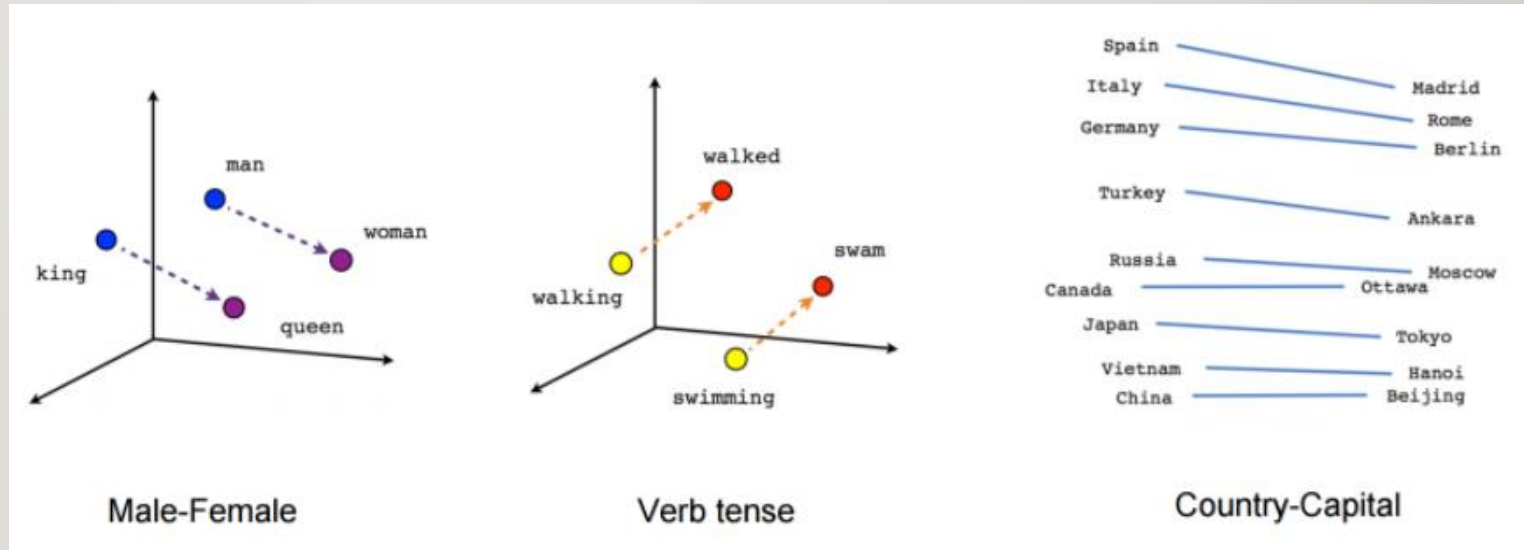
Word Embeddings

- Contextual Similarity – Captures meaning of the word enabling natural language understanding.

	Apple	Orange	Google
Fruit	0.9	0.95	0.1
Company	0.85	0.9	0.9
Classy	0.8	0.2	0.85
Network	0.12	0.8	0.2
Juice	0.87	0.92	0.05

Word Embeddings

- Visualizing the vectors on 2-dimensional space after applying dimension-reduction technique (t-SNE) shows the semantic information about words and their relationships to one another.



Embedding Projector: <http://projector.tensorflow.org/>

Natural Language Processing : Models

Basic Models:

1. ANN (Artificial Neural Network)

A narrow neural network with a single hidden layer

2. DNN (Deep Neural Network):

A deep neural network with multiple hidden layers

3. CNN (Convolution Neural Network)

Best suited for spatial data (e.g., images).

4. RNN (Recurrent Neural Network):

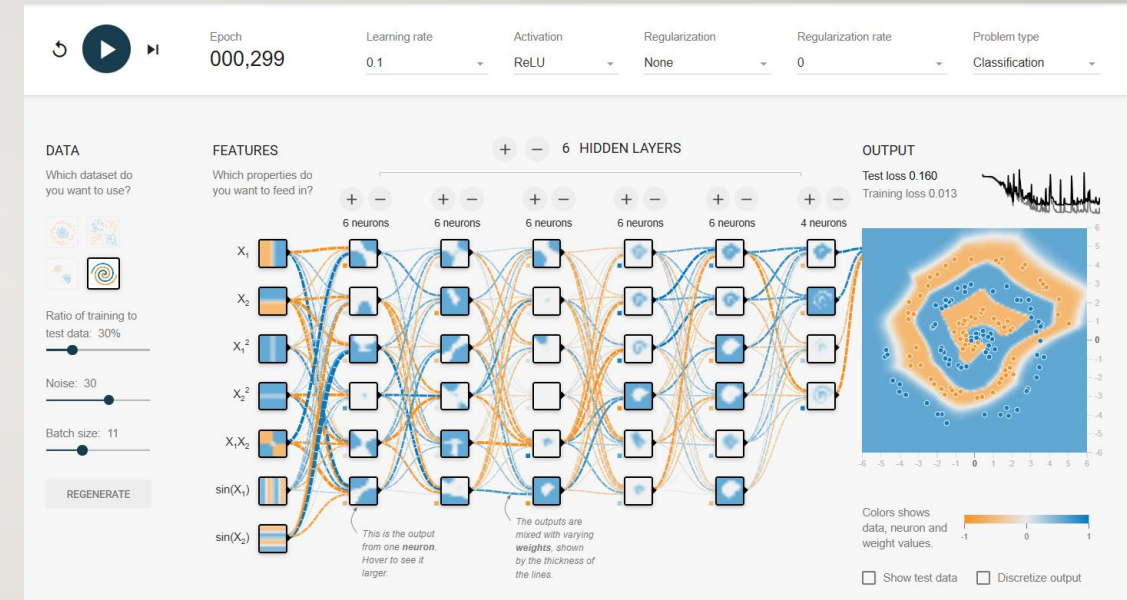
A neural network with short –term memory

5. LSTM (Long Short-term Memory):

A neural network with long-term memory

6. GRU (Gated Recurrent Unit):

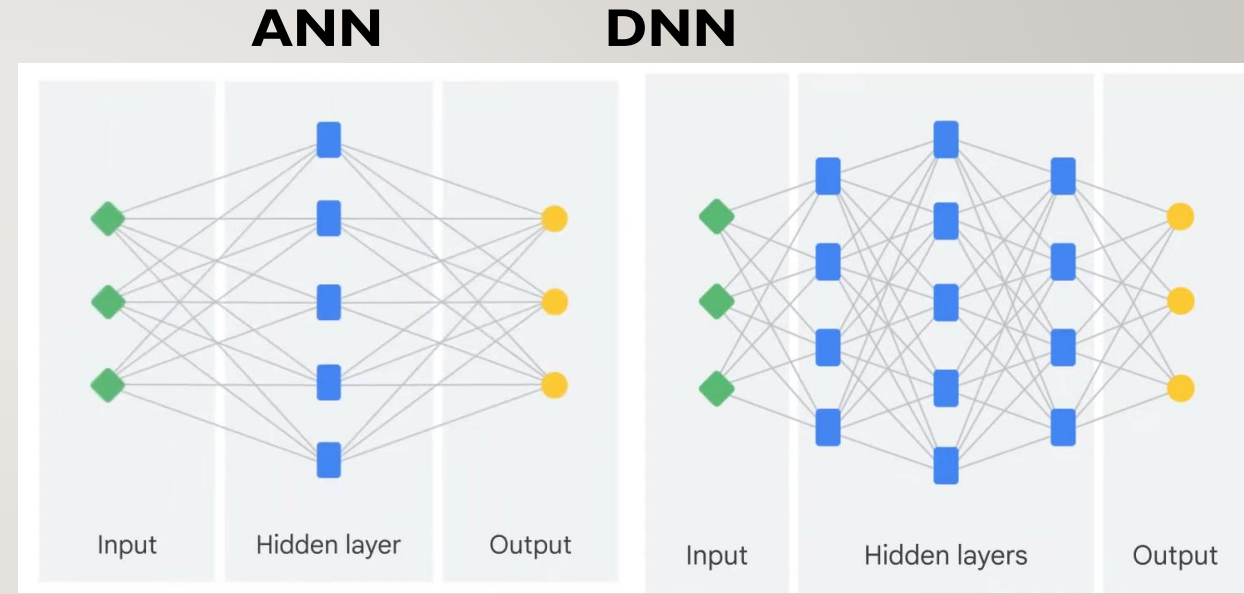
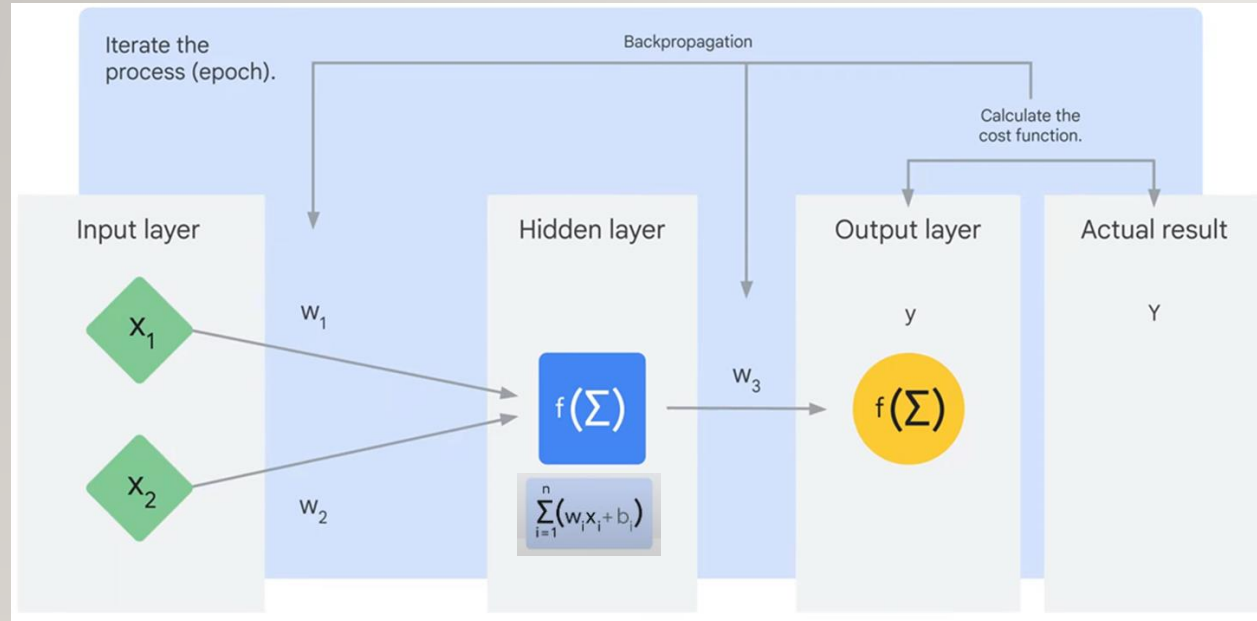
An improved and simplified variant of LSTM



Artificial neural network (ANN)

Deep neural network (DNN)

Definition: ANNs are the simplest form of neural networks, consisting of layers of interconnected neurons that process and transmit information

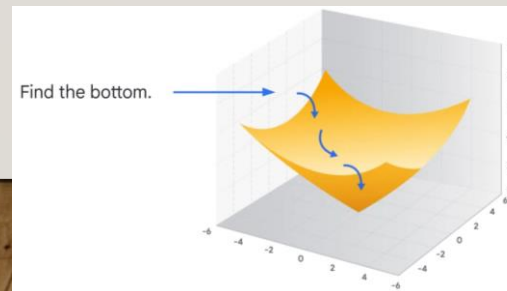


Cost Function

$$MSE = \frac{\sum_{i=1}^n (\hat{Y}_i - Y_i)^2}{n}$$

$\hat{Y}_i$ – Predicted value
 Y_i – Actual value
 n – Size of the training data

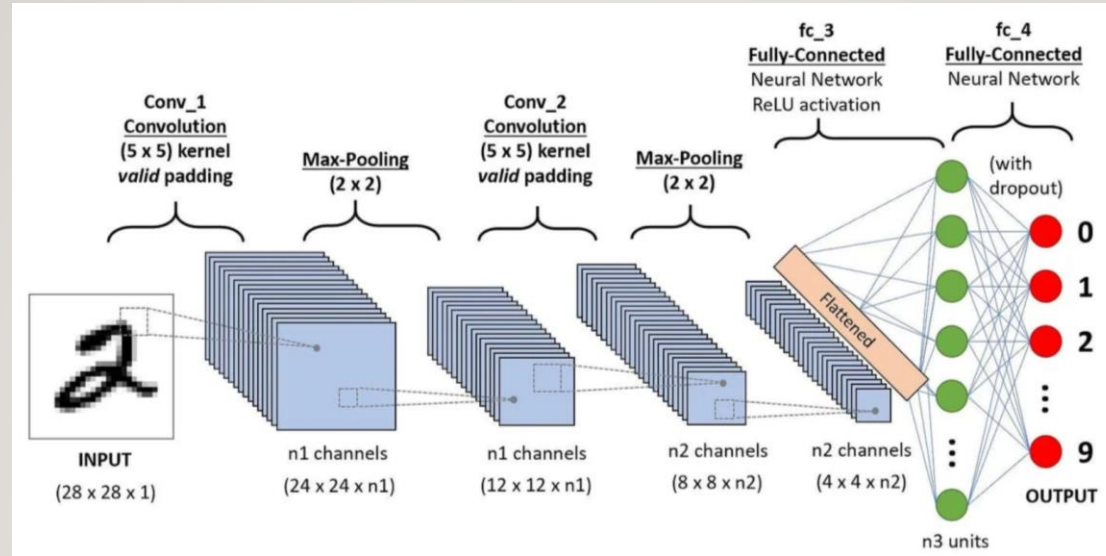
Gradient Descent



Weight Update



CNN : Convolutional Neural Networks



Definition: CNNs are specialized neural networks designed for processing structured grid data like images. They use convolutional layers to detect features.

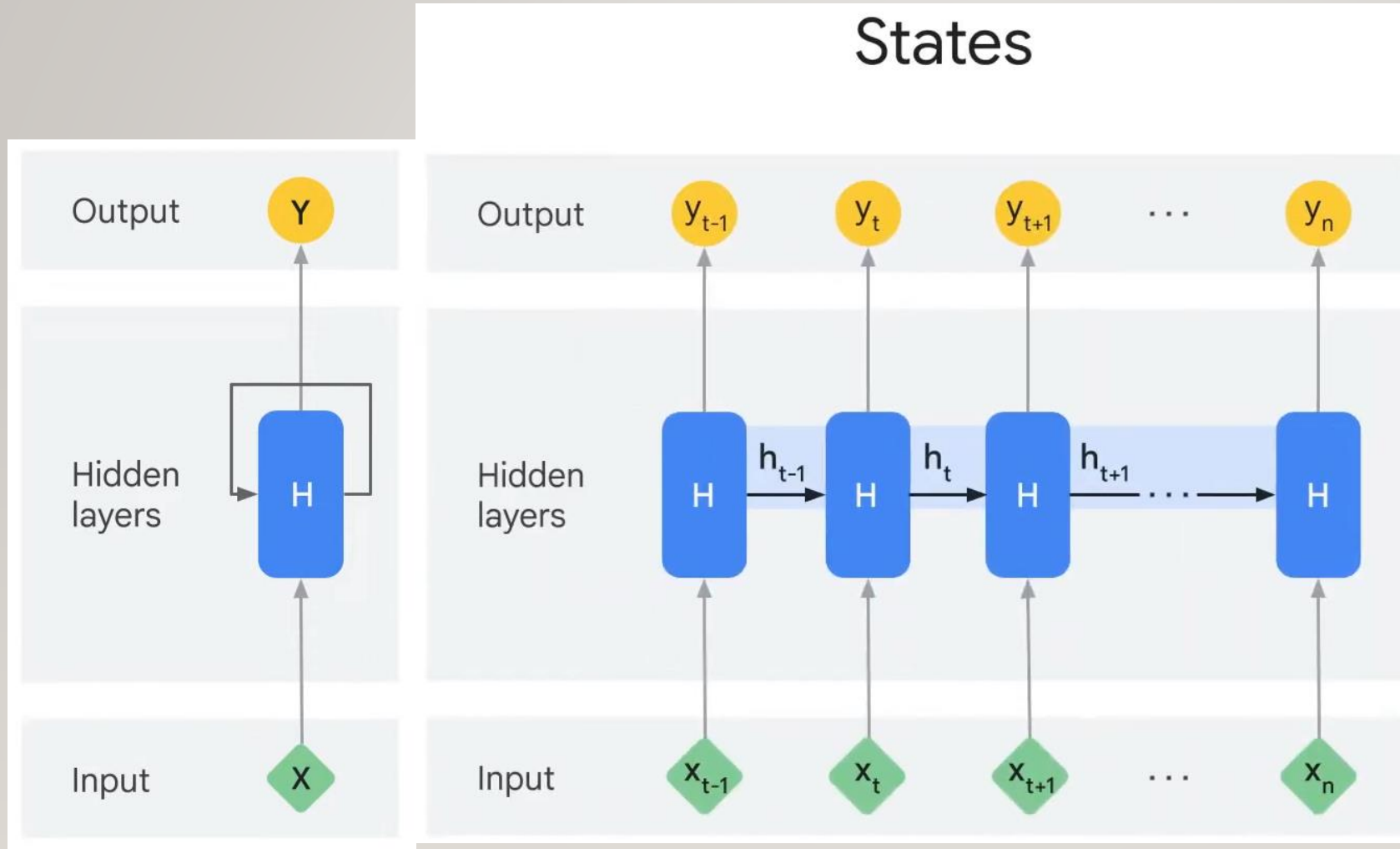
Structure:

Convolutional Layers: Apply filters to the input data to create feature maps.

Pooling Layers: Reduce the dimensionality of feature maps.

Fully Connected Layers: Connect neurons from previous layers to the output layer.

Recurrent Neural Network



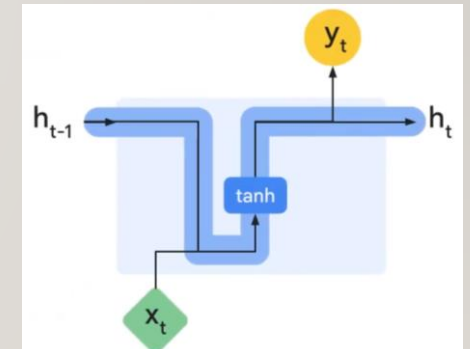
An RNN suffers from short-term memory due to vanishing gradients.

Definition: RNNs are designed for sequential data, where each neuron connects to the next layer and to neurons within the same layer.

Structure:

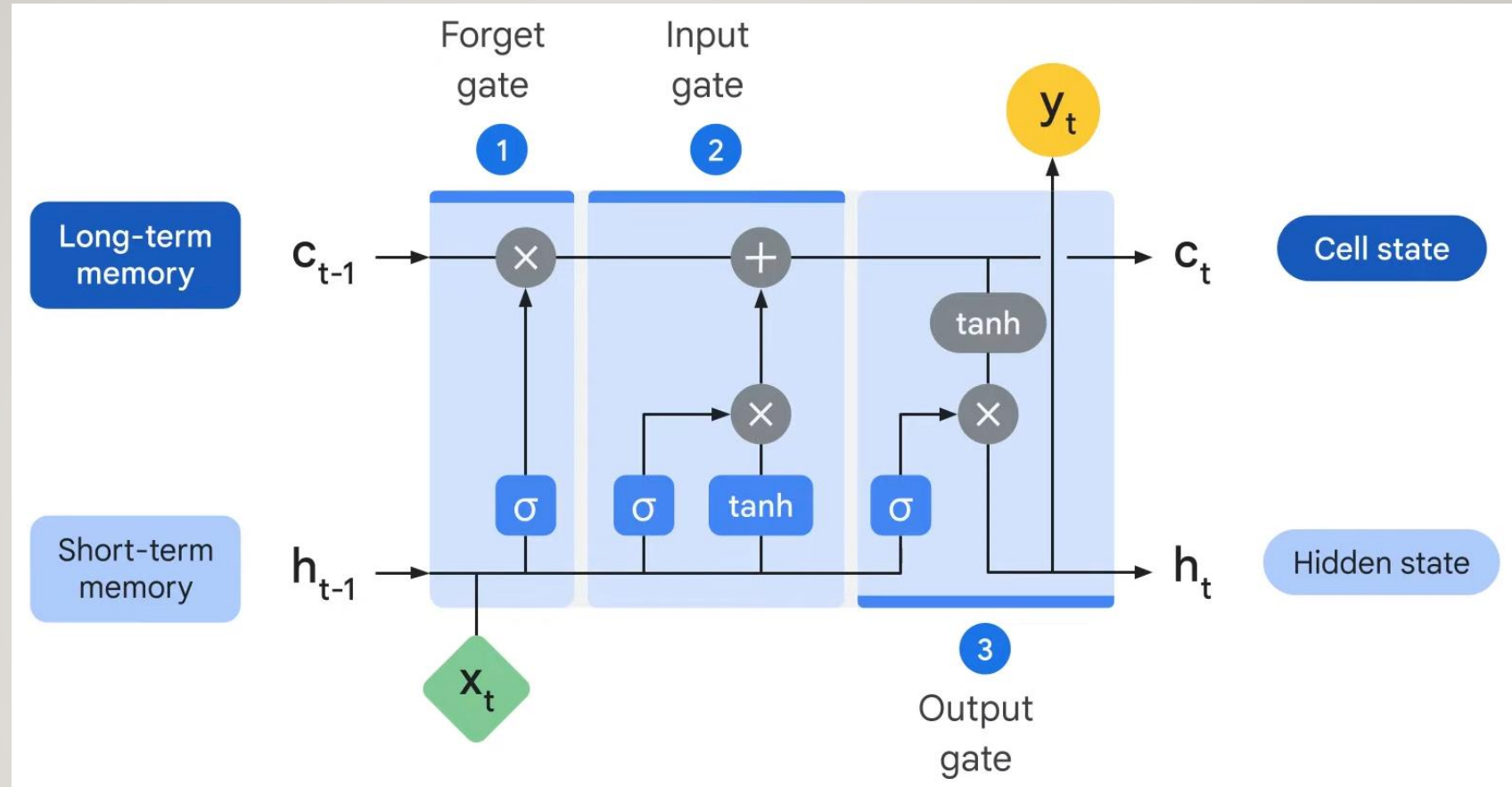
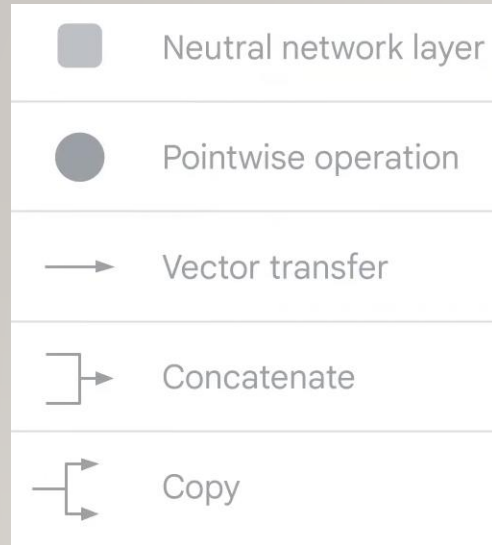
Recurrent Connections: Neurons have connections looping back to themselves, allowing information to persist.

Hidden State: Maintains a memory of previous inputs.



He grew up in the United States of America.

LSTM Cell



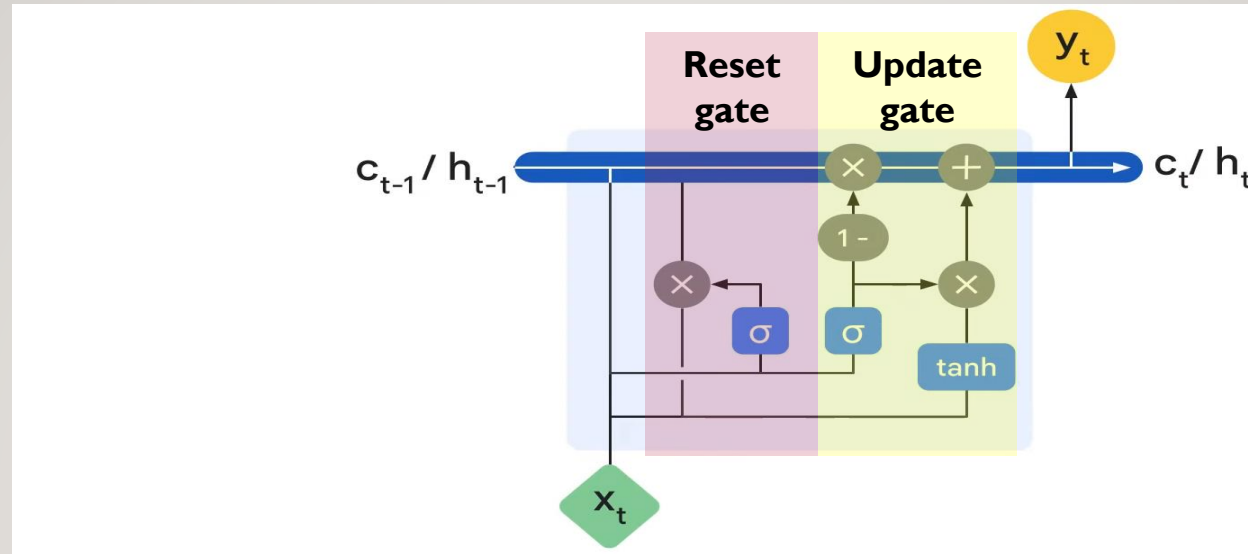
Definition: LSTMs are a type of RNN designed to learn long-term dependencies and retain information over longer sequences.

Structure:

Memory Cell: Maintains information over time.

Gates: Control the flow of information in and out of the cell (input gate, forget gate, output gate).

Gated Recurrent Unit (GRU)



Definition: GRUs are a type of RNN similar to LSTMs but with a simplified architecture, designed to capture dependencies in sequential data.

Structure:

Gates: Includes an update gate and a reset gate, simplifying the control of information flow compared to LSTMs.

Hidden State: Combines the memory cell and hidden state into a single structure.

Popular Libraries of NLP (NLTK/spaCy)

NLTK was originally created in 2001 as part of a computational linguistics course in the Department of Computer and Information Science at the **University of Pennsylvania**.

- Since then, it has been developed and expanded with the help of dozens of contributors. It has now been adopted in courses in dozens of universities, and serves as the basis of many research projects

NLTK is a leading platform for building Python programs to work with human language data.

- It provides easy-to-use interfaces to over 50 corpora and lexical resources such as WordNet, along with a suite of text processing libraries for classification, tokenization, stemming, tagging, parsing, and semantic reasoning, wrappers for industrial-strength NLP libraries.

spaCy is an open-source software library for advanced Natural Language Processing, written in the programming languages Python and Cython.

- The library is published under the MIT license and currently offers statistical neural network models for English, German, Spanish, Portuguese, French, Italian, Dutch and multi-language NER, as well as tokenization for various other languages.



Comparision Between NLTK & spaCy)

Options Vs. Best

NLTK provides a number of algorithms to choose from – good for research. Its provides different stemming libraries, for example, allow you to finely customize your model.

In contrast, spaCy implements a single stemmer, the one that the spaCy developers feel to be best (Opinionated).

Library Implementation Perspective

NLTK is essentially a string processing library, unlike spaCy which uses object-oriented approach.

Research Vs. Production

Unlike NLTK, which is widely used for **teaching and research**, spaCy focuses on providing software for **production** usage (spaCy also supports deep learning workflows that allow connecting statistical models trained by popular machine learning libraries like TensorFlow, Keras, Scikit-learn or PyTorch).



Demo Time